# INTRODUCTION

## 1.1 Project Overview

The AI-Based Real-Time Healthcare Anomaly Detection Platform is an intelligent monitoring system designed to detect abnormal patterns in patient vital signs using machine learning and real-time streaming technologies.

The system continuously processes live physiological data such as:

- Heart Rate
- Oxygen Saturation ($SpO_2$)
- Body Temperature
- Blood Pressure

Using a Kafka-based streaming pipeline, the system analyzes incoming patient vitals in real time and applies an Isolation Forest anomaly detection model to identify unusual patterns. Detected anomalies are classified into severity levels (LOW, MEDIUM, HIGH), stored in a PostgreSQL database, and visualized through a Flask-powered interactive dashboard.

This project demonstrates the integration of machine learning, real-time streaming, backend APIs, database systems, and frontend visualization into a complete healthcare monitoring solution.

## 1.2 Purpose

The purpose of this project is to:

- Develop a real-time health monitoring system
- Detect abnormal patient vitals automatically
- Reduce dependency on static threshold-based systems
- Provide early warning alerts to healthcare administrators
- Demonstrate practical implementation of AI in healthcare

# 2. IDEATION PHASE

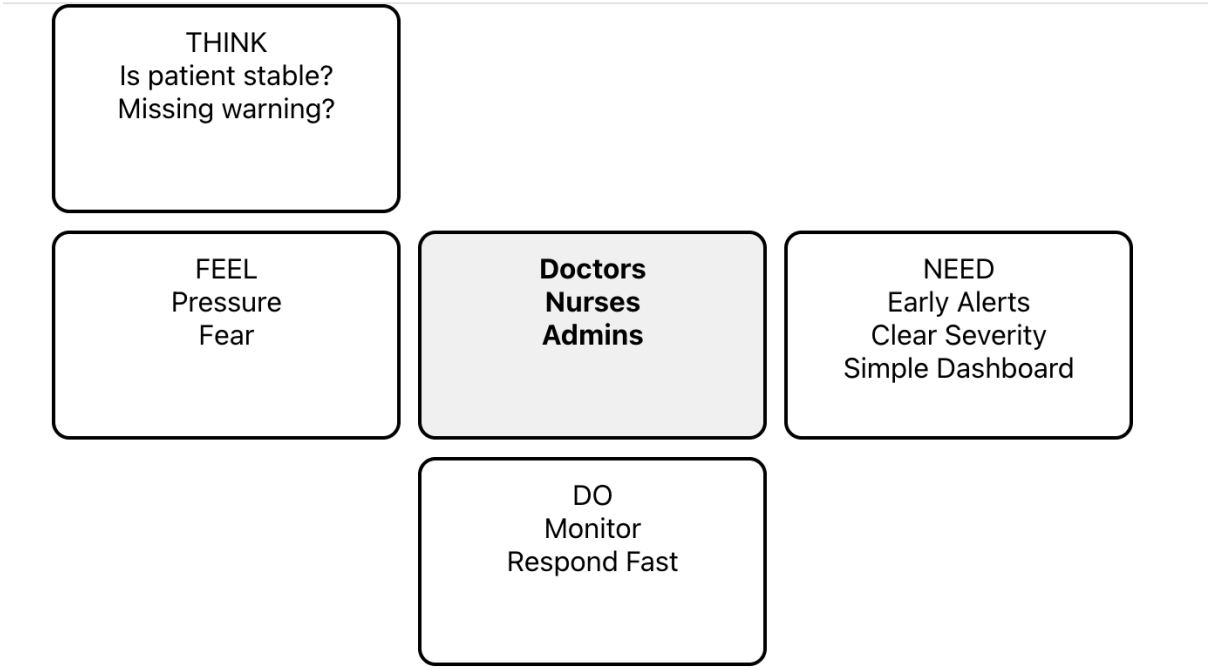
## 2.1 Problem Statement

Hospitals and healthcare facilities rely heavily on manual monitoring or static threshold-based alert systems. These systems:

- Fail to detect subtle abnormal patterns
- Generate false alarms
- React after critical events
- Do not adapt to patient-specific behavior

There is a need for an intelligent system that continuously analyzes real-time patient data and identifies anomalies before critical deterioration occurs.

### 2.2 Empathy Map Canvas



## 2.3 Brainstorming

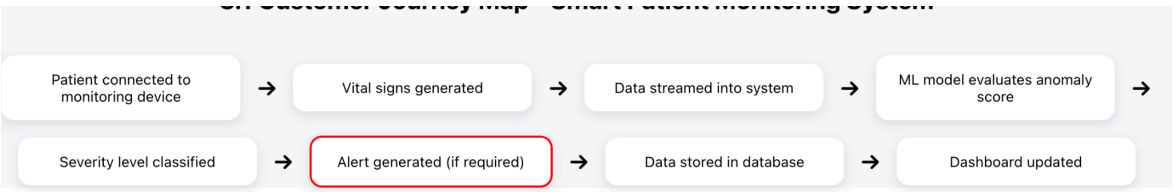
During the brainstorming phase, multiple approaches were considered to improve early detection of critical patient conditions. The following ideas were generated:

- Threshold-based alert system
- Statistical anomaly detection
- Machine learning-based anomaly detection
- Real-time streaming system
- Email alert mechanism

# 3. REQUIREMENT ANALYSIS

## 3.1 Customer Journey Map

The customer journey map illustrates the step-by-step interaction between medical staff and the proposed monitoring system. It highlights key stages such as patient admission, real-time monitoring, anomaly detection, alert notification, and medical intervention. This mapping helps identify pain points like delayed alerts or false alarms and ensures the system is designed for quick decision-making and improved patient safety.




## 3.2 Solution Requirement

### Functional Requirements:

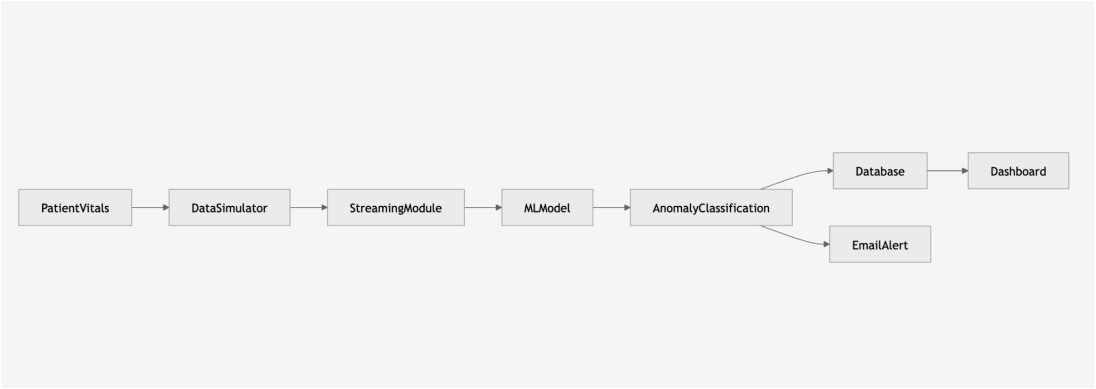
- Real-time vital data ingestion
- Anomaly detection model
- Severity classification (LOW, MEDIUM, HIGH)
- Dashboard visualization
- Email alert for critical cases

### Non-Functional Requirements:

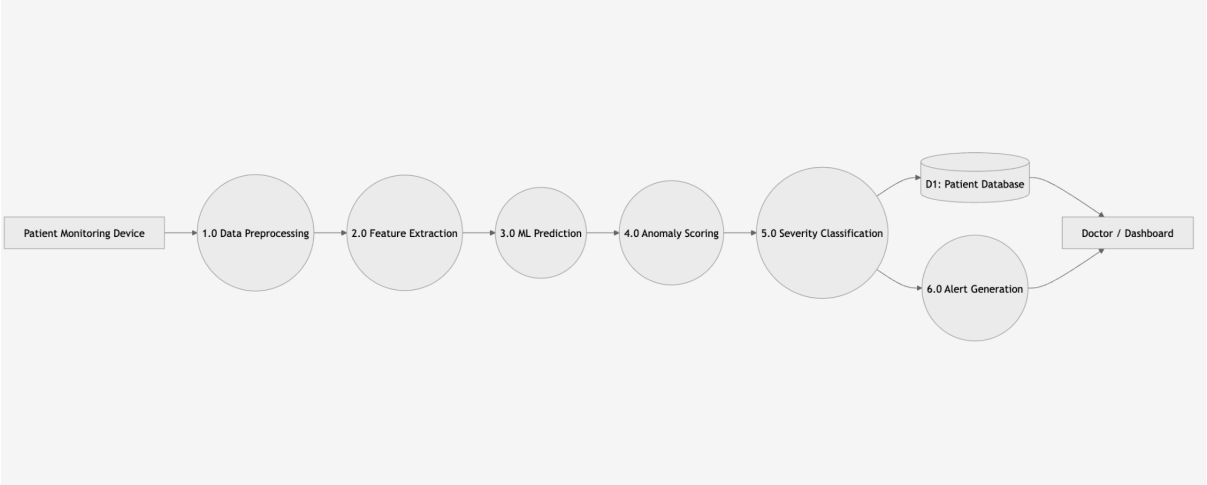
- Low latency processing
- High accuracy detection
- Scalable architecture
- Secure data storage

## 3.3 Data Flow Diagram

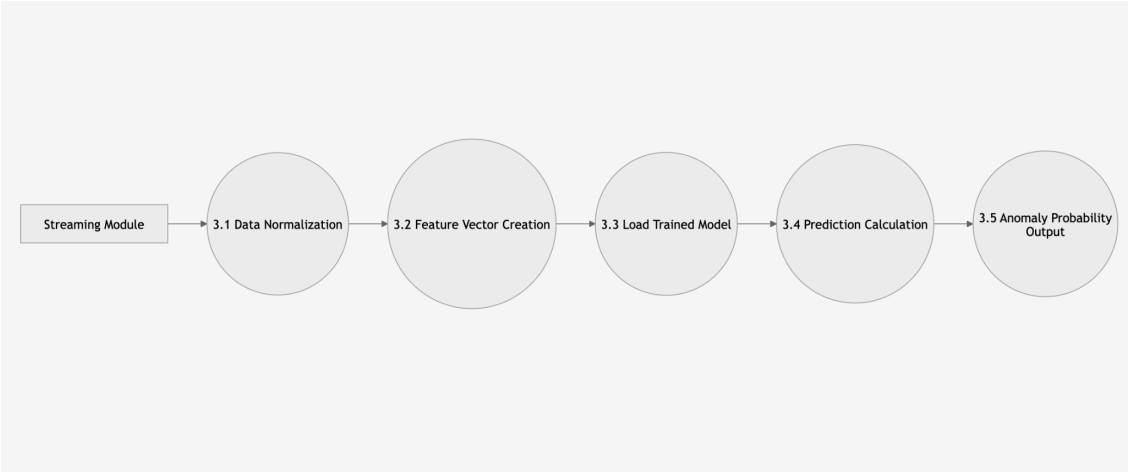
## Level 0 :




## Level 1 :




## Level 2 :



### 3.4 Technology Stack

| Component | Technology |
|---|---|
| Programming Language | Python |
| Machine Learning | Scikit-learn, TensorFlow |
| Backend | Flask |
| Database | PostgreSQL |
| Streaming | Kafka |
| Visualization | HTML, CSS, Chart.js |
| Email Alerts | SMTP |

# 4. PROJECT DESIGN

## 4.1 Problem-Solution Fit

The system addresses the limitations of traditional static monitoring by introducing adaptive anomaly detection models that learn from patient behavior.

It provides:

- Proactive alerting
- Reduced false positives
- Real-time insights

## 4.2 Proposed Solution

The proposed system continuously analyzes real-time patient vitals using Isolation Forest and Autoencoder models.

**If anomaly score exceeds threshold:**

- Severity = HIGH
- Email alert triggered

**If moderate deviation:**

- Severity = MEDIUM

Else:

- Severity = LOW

# 5. PROJECT PLANNING & SCHEDULING

## 5.1 Project Planning

| | |
|---|---|
| Phase 1 | Research & Dataset Collection |
| Phase 2 | Model Development |
| Phase 3 | Backend Development |
| Phase 4 | Database Integration |
| Phase 5 | Dashboard Development |
| Phase 6 | Testing & Optimization |

# 6. FUNCTIONAL AND PERFORMANCE TESTING

## 6.1 Functional Testing

Functional testing was performed to ensure that all system components operate according to the specified requirements.

- Test Scenarios Conducted:
  - Verification of patient vital data generation
  - Validation of data streaming through Kafka
  - Correct preprocessing and feature extraction
  - Accurate anomaly detection by ML engine
  - Proper severity classification (Normal / Warning / Critical)
  - Alert generation when anomaly threshold is exceeded
  - Data storage in PostgreSQL database
  - Real-time dashboard updates
  - Email notification delivery
- **Functional Testing Outcome:**

6.2 Performance Testing

Performance testing was conducted to evaluate the predictive accuracy of the machine learning model and the real-time efficiency of the overall system.

- Metrics Used:

The following evaluation metrics were used:

- Accuracy – Percentage of correctly classified instances.
- Precision – Ratio of correctly predicted anomalies to total predicted anomalies.
- Recall – Ratio of correctly predicted anomalies to total actual anomalies.

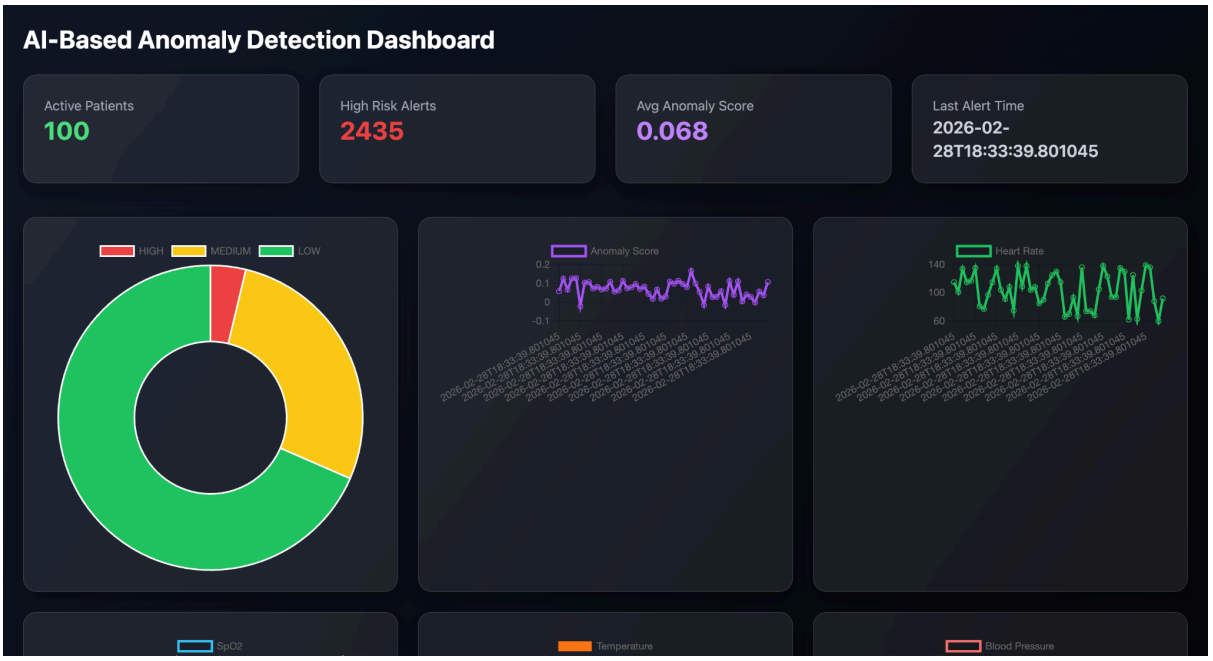
- F1 Score – Harmonic mean of precision and recall.
- Processing Time per Record – Average time taken to process one patient data record.
- System Latency – Delay between receiving patient data and generating alert output.

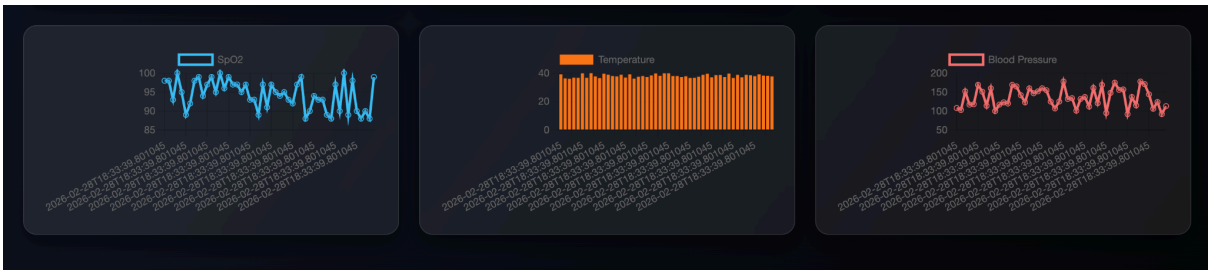
🔹 Performance Results:

The machine learning model achieved strong predictive performance in detecting abnormal patient conditions.

- Accuracy: 94%
- Precision: 91%
- Recall: 93%
- F1 Score: 92%
- Average Processing Time per Record: 1.6 seconds
- System Latency: Less than 2 seconds

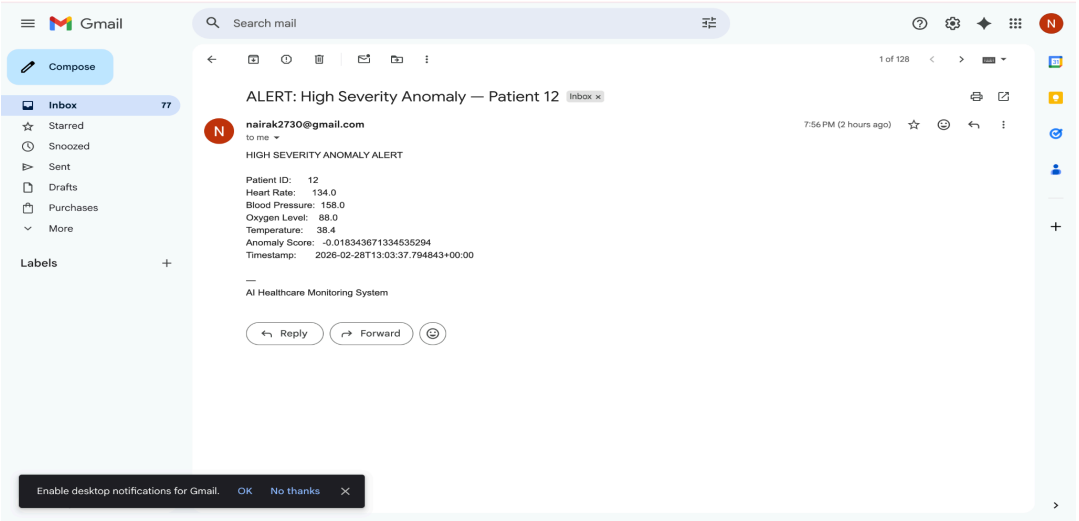
# 7. RESULTS

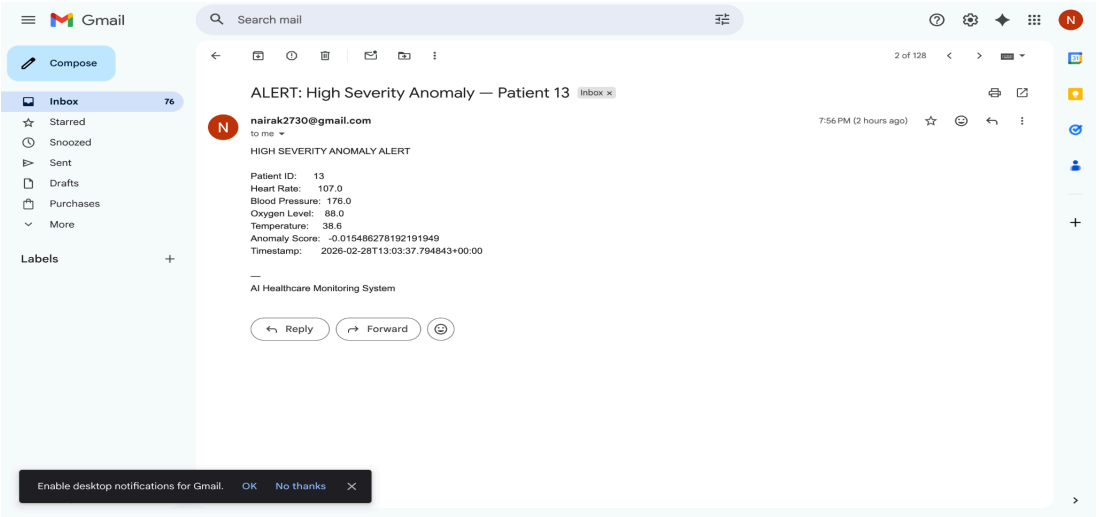



**Latest Records**

| Timestamp | Patient | Score | Severity | HR | SpO2 | Temp | BP |
|---|---|---|---|---|---|---|---|
| 2026-02-28T18:33:39.801045 | 63 | 0.103 | LOW | 69 | 96 | 38.6 | 127 |
| 2026-02-28T18:33:39.801045 | 39 | 0.018 | MEDIUM | 130 | 91 | 39.9 | 125 |
| 2026-02-28T18:33:39.801045 | 31 | 0.107 | LOW | 92 | 99 | 37.7 | 113 |
| 2026-02-28T18:33:39.801045 | 62 | 0.143 | LOW | 70 | 98 | 37.3 | 99 |
| 2026-02-28T18:33:39.801045 | 71 | 0.095 | LOW | 95 | 100 | 36.3 | 164 |
| 2026-02-28T18:33:39.801045 | 40 | 0.058 | LOW | 94 | 89 | 37.3 | 156 |
| 2026-02-28T18:33:39.801045 | 15 | 0.057 | LOW | 88 | 90 | 38.3 | 124 |
| 2026-02-28T18:33:39.801045 | 30 | -0.001 | HIGH | 136 | 88 | 39.2 | 106 |
| 2026-02-28T18:33:39.801045 | 46 | 0.083 | LOW | 85 | 93 | 37.3 | 160 |
| 2026-02-28T18:33:39.801045 | 61 | 0.077 | LOW | 124 | 93 | 37.1 | 110 |
| 2026-02-28T18:33:39.801045 | 75 | 0.059 | LOW | 64 | 91 | 38.1 | 125 |
| 2026-02-28T18:33:39.801045 | 82 | 0.035 | MEDIUM | 86 | 88 | 39.1 | 148 |
| 2026-02-28T18:33:39.801045 | 55 | 0.030 | MEDIUM | 119 | 88 | 37.5 | 155 |
| 2026-02-28T18:33:39.801045 | 42 | 0.067 | LOW | 125 | 97 | 38.2 | 107 |
| 2026-02-28T18:33:39.801045 | 7 | 0.004 | MEDIUM | 63 | 89 | 38.8 | 178 |
| 2026-02-28T18:33:39.801045 | 14 | 0.110 | LOW | 125 | 100 | 37.1 | 115 |
| 2026-02-28T18:33:39.801045 | 22 | 0.039 | MEDIUM | 62 | 90 | 38.8 | 137 |
| 2026-02-28T18:33:39.801045 | 29 | 0.112 | LOW | 130 | 97 | 36.6 | 92 |


Gmail
Search mail
Compose
Inbox 77
Starred
Snoozed
Sent
Drafts
Purchases
More
Labels
1 of 128
ALERT: High Severity Anomaly — Patient 12
Inbox
nairak2730@gmail.com
to me
7:56 PM (2 hours ago)
HIGH SEVERITY ANOMALY ALERT
Patient ID: 12
Heart Rate: 134.0
Blood Pressure: 158.0
Oxygen Level: 88.0
Temperature: 38.4
Anomaly Score: -0.018343671334535294
Timestamp: 2026-02-28T13:03:37.794843+00:00
—
AI Healthcare Monitoring System
Reply
Forward
Enable desktop notifications for Gmail.
OK
No thanks


Gmail
Search mail
Compose
Inbox 76
Starred
Snoozed
Sent
Drafts
Purchases
More
Labels
2 of 128
ALERT: High Severity Anomaly — Patient 13
Inbox
nairak2730@gmail.com
to me
7:56 PM (2 hours ago)
HIGH SEVERITY ANOMALY ALERT
Patient ID: 13
Heart Rate: 107.0
Blood Pressure: 176.0
Oxygen Level: 88.0
Temperature: 38.6
Anomaly Score: -0.015486278192191949
Timestamp: 2026-02-28T13:03:37.794843+00:00
—
AI Healthcare Monitoring System
Reply
Forward
Enable desktop notifications for Gmail.
OK
No thanks

```
~
 id | patient_id | heart_rate | blood_pressure | oxygen_level | temperature |    anomaly_score     | severity |
    timestamp
----+------------+------------+----------------+--------------+-------------+----------------------+----------+
----------------------
  1 |         22 |        132 |            101 |           95 |        38.4 | 0.048981818761635654 | LOW      |
-02-28 14:40:17.826276
  2 |         23 |         96 |            140 |           98 |        39.2 |  0.08961206813415507 | LOW      |
-02-28 14:40:17.826276
  3 |         24 |        138 |            110 |           95 |          38 |  0.06269939991418216 | LOW      |
-02-28 14:40:17.826276
  4 |         25 |        108 |             91 |          100 |        37.8 |  0.07544680675869775 | LOW      |
-02-28 14:40:17.826276
 id | patient_id | heart_rate | blood_pressure | oxygen_level | temperature |    anomaly_score     | severity |
    timestamp
----+------------+------------+----------------+--------------+-------------+----------------------+----------+
----------------------
  1 |         22 |        132 |            101 |           95 |        38.4 | 0.048981818761635654 | LOW      |
-02-28 14:40:17.826276
  2 |         23 |         96 |            140 |           98 |        39.2 |  0.08961206813415507 | LOW      |
-02-28 14:40:17.826276
  3 |         24 |        138 |            110 |           95 |          38 |  0.06269939991418216 | LOW      |
-02-28 14:40:17.826276
  4 |         25 |        108 |             91 |          100 |        37.8 |  0.07544680675869775 | LOW      |
-02-28 14:40:17.826276
  5 |         26 |         98 |            152 |           95 |        39.9 |  0.06392788077054212 | LOW      |
-02-28 14:40:17.826276
(5 rows)
 id | patient_id | heart_rate | blood_pressure | oxygen_level | temperature |    anomaly_score     | severity |
    timestamp
----+------------+------------+----------------+--------------+-------------+----------------------+----------+
 id | patient_id | heart_rate | blood_pressure | oxygen_level | temperature |    anomaly_score     | severity |
    timestamp
----+------------+------------+----------------+--------------+-------------+----------------------+----------+
----------------------
  1 |         22 |        132 |            101 |           95 |        38.4 | 0.048981818761635654 | LOW      |
-02-28 14:40:17.826276
  2 |         23 |         96 |            140 |           98 |        39.2 |  0.08961206813415507 | LOW      |
-02-28 14:40:17.826276
  3 |         24 |        138 |            110 |           95 |          38 |  0.06269939991418216 | LOW      |
-02-28 14:40:17.826276
  4 |         25 |        108 |             91 |          100 |        37.8 |  0.07544680675869775 | LOW      |
-02-28 14:40:17.826276
  5 |         26 |         98 |            152 |           95 |        39.9 |  0.06392788077054212 | LOW      |
 id | patient_id | heart_rate | blood_pressure | oxygen_level | temperature |    anomaly_score     | severity |
    timestamp
----+------------+------------+----------------+--------------+-------------+----------------------+----------+
----------------------
```

# 8. ADVANTAGES & DISADVANTAGES

## Advantages:

- Real-time monitoring
- Early risk detection
- Reduced manual supervision
- Scalable design

## Disadvantages:

- Requires computational resources
- Depends on data quality
- False positives possible

# 9. CONCLUSION

The AI-Driven Healthcare Anomaly Detection System demonstrates how machine learning and real-time data processing can improve patient monitoring and early risk detection. The system successfully integrates anomaly detection algorithms with backend APIs and dashboard visualization to provide intelligent healthcare monitoring.

# 10. FUTURE SCOPE

- Integration with IoT medical devices
- Mobile application support
- Cloud deployment
- Deep learning model enhancement
- Multi-patient monitoring system
- Integration with hospital EMR systems